

Experiment No. 7

Title

Travelling Salesman Problem Using Artificial Intelligence

Aim

To develop a Python program that solves the Travelling Salesman Problem (TSP) by finding the minimum cost path.

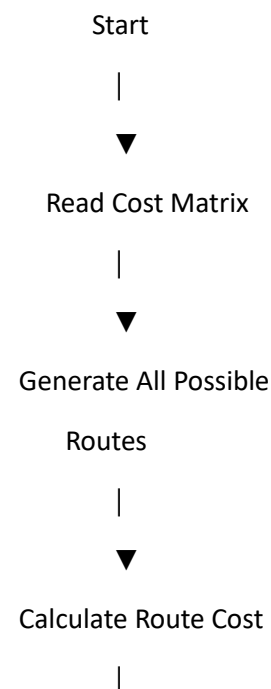
Objective

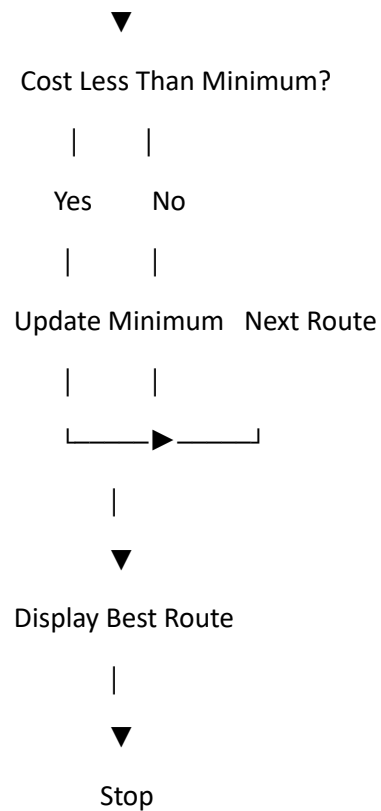
The objective of this experiment is to understand the Travelling Salesman Problem and determine the shortest possible route that visits each city exactly once and returns to the starting city.

Algorithm

- Step 1:** Define the cost matrix for all cities.
 - Step 2:** Generate all possible routes.
 - Step 3:** Calculate the total cost of each route.
 - Step 4:** Compare the route costs.
 - Step 5:** Store the route with the minimum cost.
 - Step 6:** Display the optimal route and its cost.
 - Step 7:** Stop.
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Flowchart





Python Program

```
from itertools import permutations
```

```
graph = [  
    [0, 10, 15, 20],  
    [10, 0, 35, 25],  
    [15, 35, 0, 30],  
    [20, 25, 30, 0]  
]
```

```
n = len(graph)  
cities = list(range(1, n))
```

```
min_cost = float('inf')  
best_path = None
```

```
for path in permutations(cities):  
    cost = 0  
    k = 0  
  
    for city in path:  
        cost += graph[k][city]  
        k = city  
  
    cost += graph[k][0]  
  
    if cost < min_cost:  
        min_cost = cost  
        best_path = (0,) + path + (0,)  
  
print("Best Path:", best_path)  
print("Minimum Cost:", min_cost)
```

Sample Output

Best Path: (0, 1, 3, 2, 0)

Minimum Cost: 80

Result

The minimum-cost route for the Travelling Salesman Problem was successfully obtained.

Conclusion

Thus, the Travelling Salesman Problem was successfully solved by evaluating all possible routes and selecting the one with the minimum travel cost.